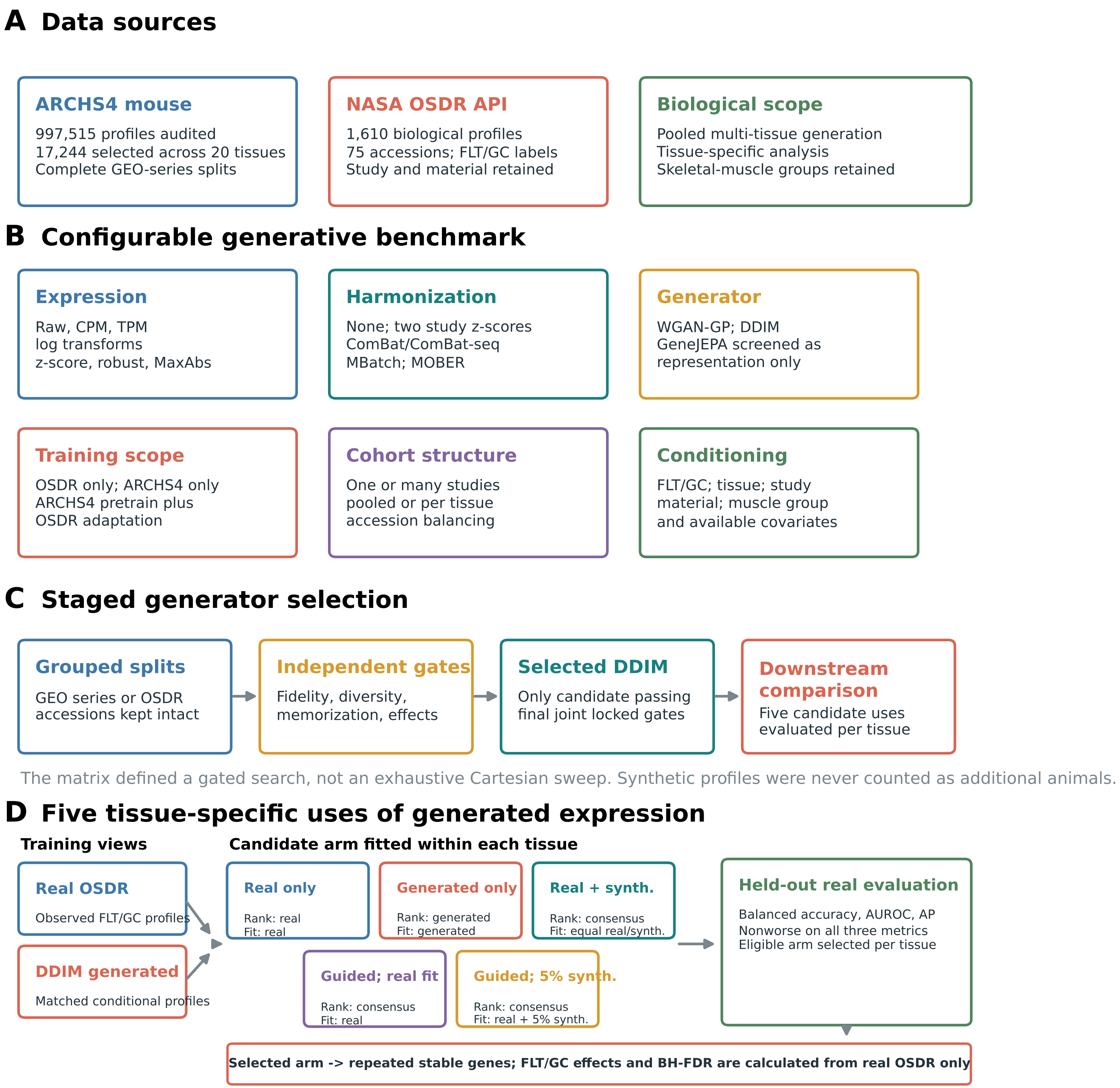

A Data sources
ARCHS4 mouse
997,515 profiles audited
17,244 selected across 20 tissues
Complete GEO-series splits
NASA OSDR API
1,610 biological profiles
75 accessions; FLT/GC labels
Study and material retained
Biological scope
Pooled multi-tissue generation
Tissue-specific analysis
Skeletal-muscle groups retained
B Configurable generative benchmark
Expression
Raw, CPM, TPM
log transforms
z-score, robust, MaxAbs
Harmonization
None; two study z-scores
ComBat/ComBat-seq
MBatch; MOBER
Generator
WGAN-GP; DDIM
GeneJEPA screened as
representation only
Training scope
OSDR only; ARCHS4 only
ARCHS4 pretrain plus
OSDR adaptation
Cohort structure
One or many studies
pooled or per tissue
accession balancing
Conditioning
FLT/GC; tissue; study
material; muscle group
and available covariates
C Staged generator selection
Grouped splits
GEO series or OSDR
accessions kept intact
Independent gates
Fidelity, diversity,
memorization, effects
Selected DDIM
Only candidate passing
final joint locked gates
Downstream
comparison
Five candidate uses
evaluated per tissue
The matrix defined a gated search, not an exhaustive Cartesian sweep. Synthetic profiles were never counted as additional animals.
D Five tissue-specific uses of generated expression
Training views
Candidate arm fitted within each tissue
Real OSDR
Observed FLT/GC profiles
DDIM generated
Matched conditional profiles
Real only
Rank: real
Fit: real
Generated only
Rank: generated
Fit: generated
Real + synth.
Rank: consensus
Fit: equal real/synth.
Guided; real fit
Rank: consensus
Fit: real
Guided; 5% synth.
Rank: consensus
Fit: real + 5% synth.
Held-out real evaluation
Balanced accuracy, AUROC, AP
Nonworse on all three metrics
Eligible arm selected per tissue
Selected arm -> repeated stable genes; FLT/GC effects and BH-FDR are calculated from real OSDR only